

Factor and Categorical Matrices

In this lecture we will discuss the **factor()** function and its use for creating categorical matrices. This specific function will become extremely useful when we begin to apply data analysis and machine learning techniques to our data, this idea is sometimes also known as creating *dummy variables*.

Let's start by showing an example of why and how we build this matrix. Imagine we have the following vectors representing data from an animal sanctuary for dogs ('d') and cats ('c') where they each have a corresponding id number in another vector.

In [6]:

```
animal <- c('d','c','d','c','c')
id <- c(1,2,3,4,5)
```

We want to convert the animal vector into information that an algorithm or equation can understand more easily. Meaning we want to begin to check how many categories (factor levels) are in our character vector.

We can pass the vector through the **factor()** function like so to get this information:

In [9]:

```
factor ani <- factor(animal)
```

In [10]:

```
# Will show levels as well on RStudio or R Console
factor ani
```

Out[10]:

```
 d  c  d  c  c
```

We can see that we have two levels, 'd' and 'c'. In R there are two distinct types of categorical variables, a **ordinal categorical variable** and a **nominal categorical variable**.

Nominal categorical variables don't have any order, such as dogs and cats (there is no order to them). Versus Ordinal categorical variables (as the name implies) do have an order. For example, if you had the vector:

In [13]:

```
ord.cat <- c('cold','med','hot')
```

You could begin to assign in order to these variables, such as:

```
cold < med < hot
```

if you wanted to assign an order while using the **factor()** function, you can pass in the arguments **ordered=True** and then pass in the **levels=** and pass in a vector in the order you want the levels to be in. So for example:

In [20]:

```
temps <- c('cold','med','cold','med','hot','hot','cold')
fact.temp <- factor(temps,ordered=TRUE,levels=c('cold','med','hot'))
fact.temp
```

Out[20]:

```
 cold  med  cold  med  hot  hot  cold
```

This information is useful when used along with the **summary()** function which is an amazingly convenient function for quickly getting information from a matrix or vector. For example:

In [21]:

```
summary(temps)
```

Out[21]:

Length	Class	Mode
7	character	character

In [22]:

```
summary(fact.temp)
```

Out[22]:

cold
3
med
2
hot
2

Later on we will revisit this idea of using the **factor()** function and **ordered** argument. For now that is all we need to know.